

ELECTROMYOGRAPHY

What is electromyography?

- **Electromyography (EMG)** is a technique used to measure the electrical activity produced by skeletal muscles.
- When your brain sends a signal to contract a muscle, **motor neurons** activate groups of muscle fibres called **motor units**.
- The activation of muscle fibres generates tiny electrical signals known as **muscle action potentials**.
- **Surface EMG** records these signals using electrodes placed on the skin above the muscle.
- The strength of the EMG signal generally increases as more motor units are recruited and muscles produce greater force.
- EMG does **not measure force directly**; it measures the electrical activity associated with muscle activation.
- EMG signals are typically only a few **microvolts to millivolts** in amplitude and must be amplified and processed before analysis.
- Signal-processing techniques such as **filtering**, **rectification**, and **smoothing** are often used to extract useful information from raw EMG data.

Why is EMG useful?

- EMG is widely used in **biomechanics**, **rehabilitation**, **sports science**, **prosthetics**, and **human-computer interaction**.
- It provides insight into how the nervous system controls movement.
- EMG can be used as an intuitive control signal for assistive devices, virtual reality systems, and interactive games.



How to set up this demo:

- Plug the green AUX cable into the spikerbox. Use the end that is labelled with “Spikerbox”. The other end labelled “Smartphones” goes to the iPad (with the USB-C adapter)
- Plug the orange cable into Channel 1 and make sure the switch next to it is on “Ch1”
- Turn the spikerbox on by rotating the volume wheel. You should here a “click” when its on. You don’t need to turn the volume up a lot, but it can be interesting to hear the sound of the muscles as well.
- Place two electrodes on the biceps brachii muscle of the student, and one electrode on the elbow bone. The two electrodes on the muscles should be aligned with the muscle fibre orientation.
- Connect the muscle electrodes with the red cables and the bone electrode with the black cable.

